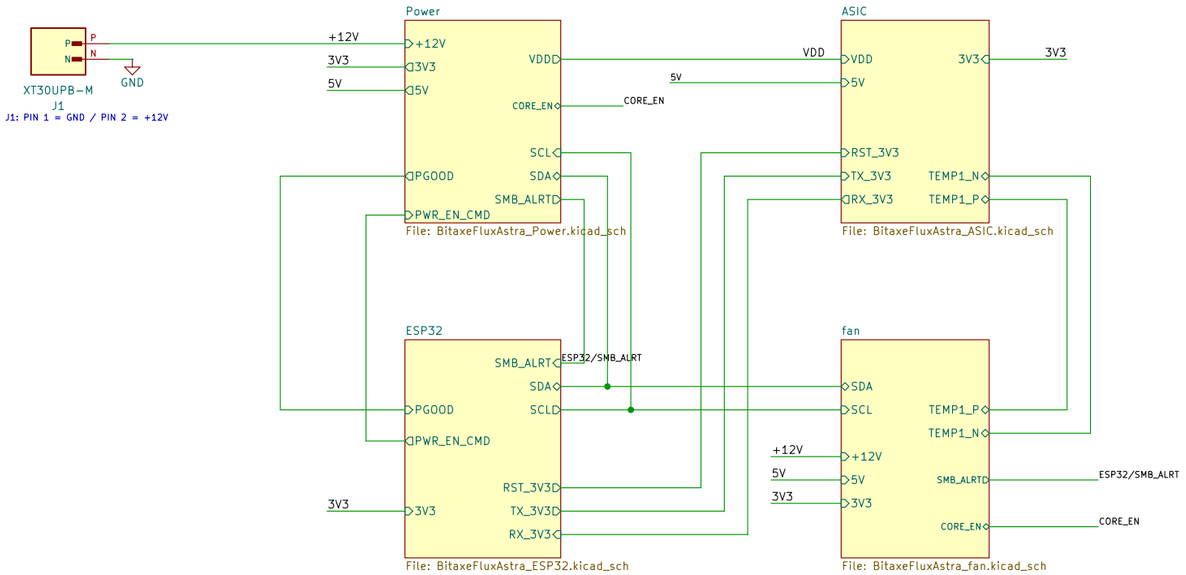
12V INPUT PROTECTION TO BE QUALIFIED BEFORE FABRICATION RELEASE



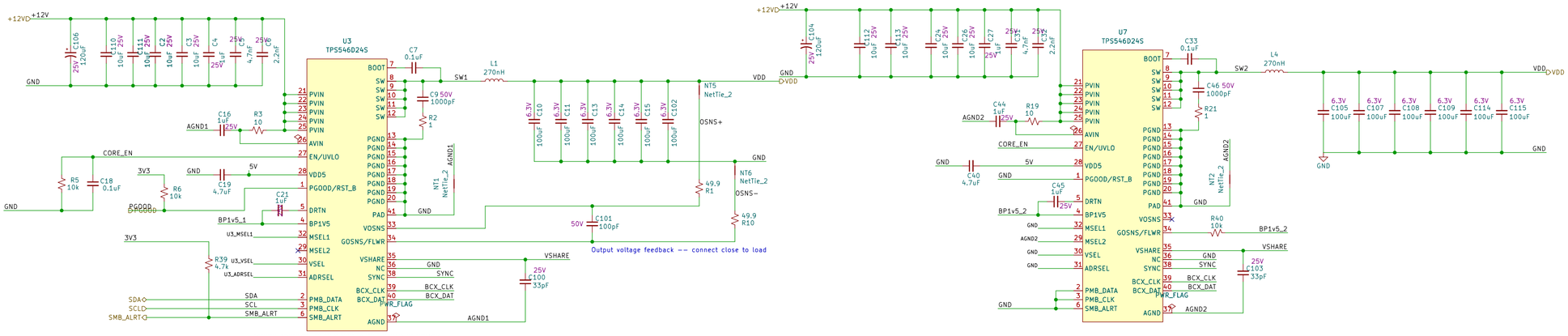
A1 ENGINEERING – FIRST ARTICLE SCHEMATIC REVIEW
Open qualification items are listed in A0_to_A1_Schematic_Changes.md.

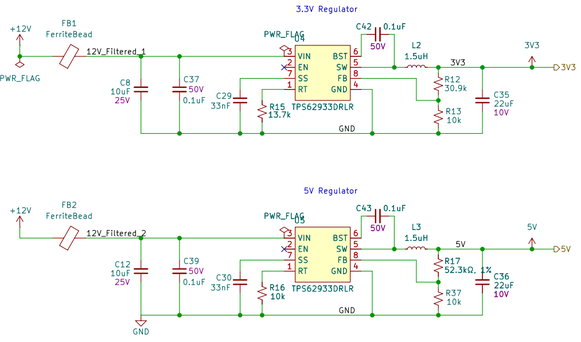
A1 ENGINEERING – FIRST ARTICLE SCHEMATIC REVIEW
bitaxeorg

Sheet: /
File: BitaxeFluxAstra.kicad_sch

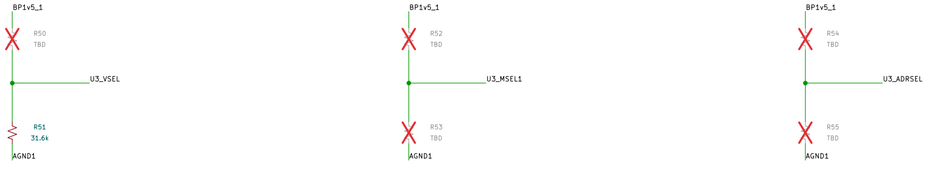
Title: Bitaxe Flux Astra

Size: A3	Date: 2026-09-20	**Rev: RevA**
KiCad E.D.A. 10.0.5		Id: 1/5





A1: Bitmain BM1373. Part Number: BM1373. Configuration: Mode 1.
Approved pin mapping, Mode strap and RI termination preserved.
ENGINEERING DRAFT: power startup, thermal protection and mechanical stack require validation.



L1 / L4 = 270nH SLC1175-271MEC
A1 FIRST-ARTICLE VALUE — LOOP / CURRENT QUALIFICATION REQUIRED

A1 ENGINEERING — FIRST ARTICLE SCHEMATIC REVIEW
bitaxeorg
Sheet: /Power/
File: BitaxeFluxAstra_Power.kicad_sch
Title: Bitaxe Flux Astra

Size: A2	Date: 2026-09-20	**Rev: RevA**
KiCad E.D.A. 10.0.5		Id: 2/5

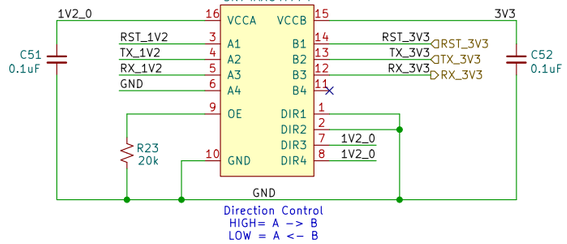
USB_VBUS: HOLD
VBUS_MONITOR_REQUIRED_IF_TINYUSB_SELF_POWERED_MODE_IS_USED
NO CONNECTION TO BOARD 5V / 3V3. FIRMWARE MODE NOT ESTABLISHED.

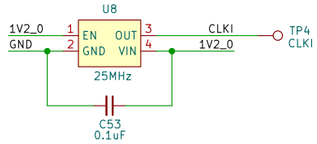
J2: pins 19 / 26 / 27 / 28 / 29 intentional NC.
FPC mating side / final suffix: HOLD: original pin order retained.

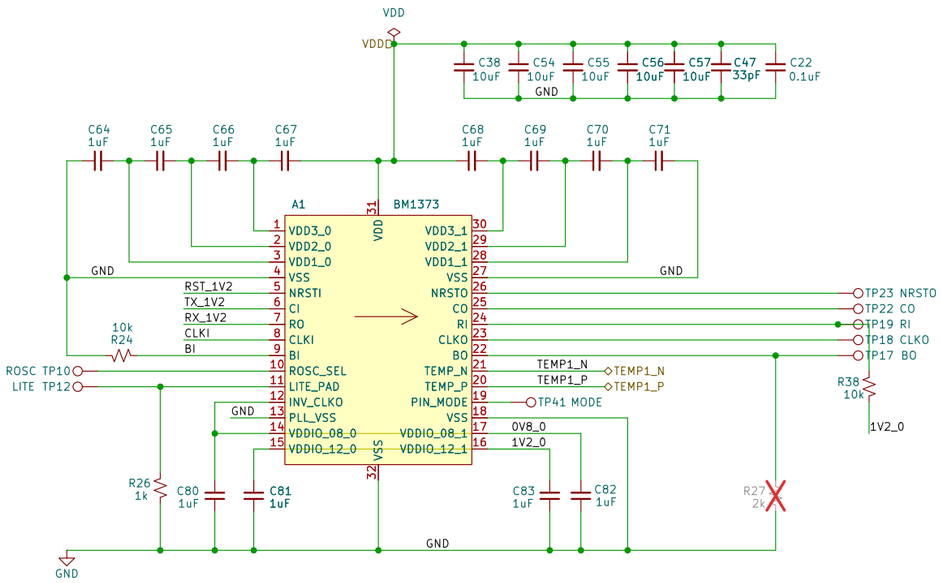
A1 ENGINEERING – FIRST ARTICLE SCHEMATIC REVIEW	
bitaxeorg	
Sheet: /ESP32/	
File: BitaxeFluxAstra_ESP32.kicad_sch	
Title: Bitaxe Flux Astra	
Size: A3	Date: 2026-09-20
KiCad E.D.A. 10.0.5	Id: 3/5

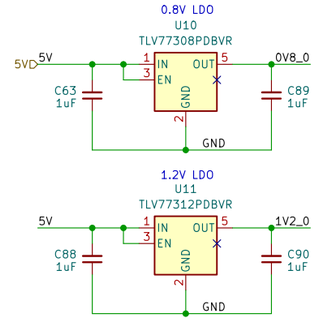
Rev: RevA

A1: Bitmain BM1373. Part Number: BM1373. Configuration: Mode 1.
Approved pin mapping, Mode strap and RI termination preserved.
ENGINEERING DRAFT: power startup, thermal protection and mechanical stack require validation.









PIN 19: DATASHEET/REFERENCE INTENTIONAL FLOAT
MODE 1 COMMUNITY IMPLEMENTATION: TP41 ONLY
PIN 10: ROSC_SEL REFERENCE FLOAT / TEST POINT: NO BIAS ADDED
PINS 22/23/25/26: UNLOADED TEST OUTPUTS: R27 DNP

U2: OE LOW = ENABLED; OE HIGH = HIGH IMPEDANCE
R23 20k PULLDOWN RETAINED: B4 INTENTIONAL NC
U8: 25MHz / 1V2_0: EN TIED HIGH. EXACT ORDER CODE QUALIFICATION HOLD.
U10 / U11 PIN 4: TRUE NC / INTENTIONAL_NC.

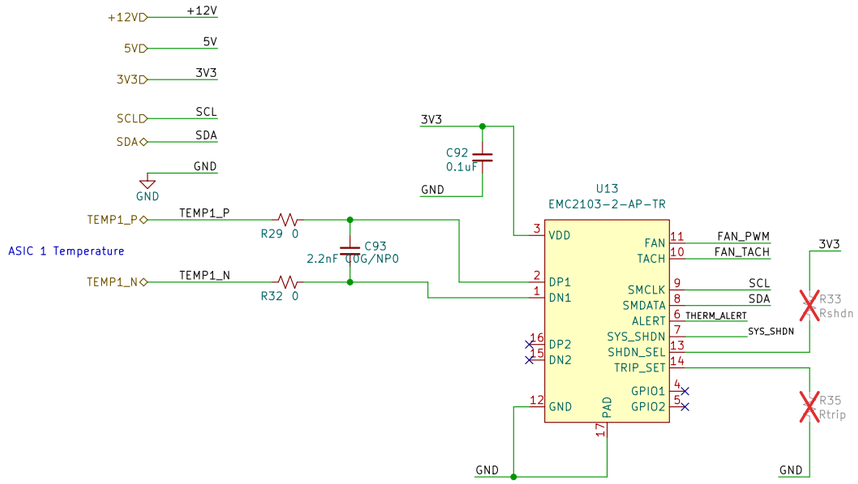
A1 ENGINEERING – FIRST ARTICLE SCHEMATIC REVIEW
bitaxeorg

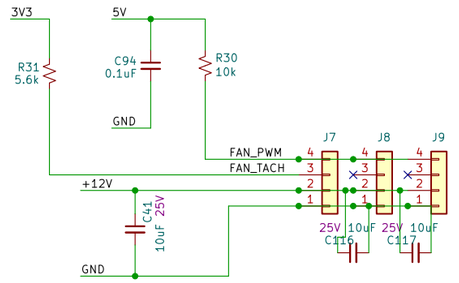
Sheet: /ASIC/
File: BitaxeFluxAstra_ASIC.kicad_sch

Title: Bitaxe Flux Astra

Size: A3	Date: 2026-09-20	**Rev: RevA**
KiCad E.D.A. 10.0.5		Id: 4/5

EXTERNAL DIODE 2 UNUSED — MUST REMAIN DISABLED/MASKED IN FIRMWARE





J7: CPU fan with tach feedback. J8/J9: shared PWM; no tach.
Firmware must mask unused diode 2 and handle diode/fan faults.







HARDWARE SHUTDOWN CIRCUIT PROVIDED — TRIP THRESHOLD NOT YET QUALIFIED
R33 / R35 DNP. D1 ANODE = CORE_EN; CATHODE = SYS_SHDN.
R48 1k ON POWER SHEET LIMITS GPIO CONTENTION; VERIFY CLAMP LOW MARGIN.

A1 ENGINEERING — FIRST ARTICLE SCHEMATIC REVIEW
bitaxeorg
Sheet: /fan/
File: BitaxeFluxAstra_fan.kicad_sch
Title: Bitaxe Flux Astra

Size: A3	Date: 2026-09-20	**Rev: RevA**
KiCad E.D.A. 10.0.5		Id: 5/5